Industrial System Analysis for Monitoring Alert Prediction

Introduction

This report summarizes a comprehensive data science workflow implemented in a Jupyter Notebook to predict monitoring alerts for an industrial well. The analysis includes data exploration, preprocessing, feature engineering, model development, and evaluation.

Part 2 (open-ended) includes findings for anomaly detection, aiming to provide a robust predictive maintenance system.

1. Data Exploration and Preprocessing

1.1. Data Description: The dataset comprises time-series sensor readings and operational metrics from a single industrial well (ID 101) collected over several months in 2023. Key sensor data includes ambient and surface temperature, various pressure readings, motor power, and vibration metrics.

1.2. Exploratory Analysis:

- **Distribution Insights:** Analysis of sensor value distributions revealed distinct patterns. For instance, acoustic sensor readings approximate a normal distribution, while casing pressure and tubing pressure exhibit significant skewness with notable outliers.
- **Temporal Trends:** Time-series analysis indicated that alerts tend to occur in bursts with unique cyclic behavior. Seasonal trends were also observed in temperature-related features.
- **Missing Value Assessment:** A thorough examination of missing values identified features requiring specific handling. **Operational notes**, **lubrication_status** and **maintenance codes** were dropped due to high missingness.

1.3. Preprocessing Steps:

- Relevant time-based features (year, month, day, weekday, etc.) were extracted to enrich the dataset for predictive modeling.
- Categorical variables, such as well operating status, were transformed using one-hot encoding since it had only 4 unique values.
- Sensor data was scaled using the **RobustScaler** to minimize the impact of outliers.
- Lagged features were engineered to capture temporal dependencies within the sensor readings.

2. Modeling for Monitoring Alert Prediction

2.1. Baseline and Model Selection: A DummyClassifier served as the baseline model. A RandomForestClassifier was subsequently chosen for its ability to handle imbalanced data and capture non-linear relationships.

2.2. Class Imbalance Handling: To address the inherent imbalance between normal operations and alert events, class weight adjustments were implemented within the RandomForestClassifier.

2.3. Performance Metrics: The model achieved the following performance on the validation set:

Metric	Value
Accuracy	98.55%
Precision	99.54%
Recall	85.08%
ROC AUC	0.998

These metrics indicate excellent overall performance, with a particularly high precision for predicting alert conditions. ***However, the recall score suggests room for improvement in capturing all actual alert events***.

2.4. Model Improvement after Hyperparameter Tuning: Our goal is to improve recall since missing an actual Alert event can be critical. Keeping that in mind, we improved the model **recall (~88%)** by performing Hyperparameter Tuning using GridSearchCV

2.5. Threshold Tuning: Recognizing the importance of the balance between precision and recall in an industrial monitoring context, a detailed analysis of the F1 score across various classification thresholds was conducted. This allowed for the selection of an optimal cutoff point tailored to the desired trade-off.

2.6. Feature Importance: Both Gini-based and permutation-based feature importance analyses provided valuable insights into the drivers of alert predictions:

- Immediate sensor readings (e.g., surface temperature and motor power), along with their short-term lagged values, were identified as critical predictors.

- Temporal features, such as the day of the year and the day of the month, play a significant role in determining alert conditions, likely reflecting operational schedules or environmental cycles.
- Permutation analysis highlighted seismic activity as a significant predictor, suggesting its potential influence on well operations and alert triggers.

PART 2

Anomaly Detection with Isolation Forest

Objective: To complement the alert prediction model, an anomaly detection system was developed to provide an early warning of unusual operational conditions that might precede critical monitoring alerts.

Implementation: An Isolation Forest algorithm was trained on the processed training data. Anomaly scores were calculated for each data point, and an initial threshold (the 95th percentile of anomaly scores in the training data) was used to flag potential anomalies.

Performance and Improvement: Initial results showed high precision in identifying normal operations but also a tendency to miss a significant number of anomalies (high false negative rate). To address this, follow-up efforts focused on threshold tuning and hyperparameter optimization using GridSearchCV with a custom F1 scoring function specifically designed for anomaly detection. These improvements led to a higher recall for anomalous events, enhancing the system's ability to detect deviations from normal behavior.

3. Conclusions and Future Directions

3.1. Overall Outcome: The developed RandomForestClassifier demonstrates high precision in predicting monitoring alerts, achieving excellent performance on the validation set. While the recall could be further improved through more refined threshold optimization, the current balance offers a strong foundation for proactive maintenance.

3.2. Anomaly Detection: The enhanced Isolation Forest model shows significant promise as an early warning system. Ongoing tuning to optimize the trade-off between overall accuracy and improved anomaly recall is crucial, particularly in high-risk industrial environments where missing an anomaly can have substantial consequences.

3.3. Future Work: To further enhance the system, the following future work is recommended:

- **Ensemble Methods:** Explore the integration of complementary anomaly detection techniques, such as Autoencoders, to capture a broader range of anomalous patterns.
- **Feature Explainability:** Implement methods for providing per-sample feature importance explanations to better understand the specific factors contributing to individual anomaly detections. This can aid in diagnosing the root causes of anomalies.
- **Real-Time Monitoring:** Transition the developed models towards a real-time monitoring system capable of continuously processing streaming sensor data and providing immediate alerts based on predicted probabilities and anomaly scores.